

AI Important Questions with Answers (Theory + Example + Numerical)

1. Problem Space

Theory (English): Problem space means the complete set of all possible states from initial state to goal state in an AI problem. It contains states, operators, initial state and goal state.

Samajhne ke liye (Hindi): Problem space matlab kisi problem ko solve karne ke liye saare possible states ka collection.

Example: In 8-puzzle, each board arrangement is a state. Moving blank tile left/right/up/down are operators.

2. Informed vs Uninformed Search

Theory (English): Uninformed search uses no extra knowledge and explores blindly. Informed search uses heuristic information to reach goal faster.

Hindi: Uninformed search bina guidance ke search karta hai, informed search heuristic use karta hai.

Example: BFS is uninformed, A* is informed.

3. BFS (Breadth First Search)

Theory (English): BFS explores level by level and uses queue data structure.

Hindi: BFS pehle nearest nodes visit karta hai phir next level par jata hai.

Example: A->B,C ; B->D,E ; traversal = A,B,C,D,E.

4. A* Algorithm

Theory (English): A* selects node with minimum $f(n)=g(n)+h(n)$, where g is path cost and h is heuristic estimate.

Hindi: A* me actual cost aur estimated cost dono use hote hain.

Numerical Example: If node B has $g=4$ and $h=3$, then $f(B)=7$. If node C has $g=5$ and $h=1$, then $f(C)=6$, so choose C.

5. Hill Climbing

Theory (English): Hill climbing is a local search algorithm that moves toward better state step by step.

Hindi: Current state se better neighbour choose karte hain jab tak best state na mil jaye.

Example: Highest value neighbour is selected at each step.

6. Crypt Arithmetic: SEND + MORE = MONEY

Theory (English): Each letter represents unique digit and constraints must satisfy addition.

Hindi: Har letter alag digit hota hai aur addition correct hona chahiye.

Example Solution:

SEND = 9567

MORE = 1085

MONEY = 10652

7. MiniMax

Theory (English): MiniMax is used in game playing where MAX player maximizes score and MIN player minimizes score.

Hindi: Ek player best maximum choose karta hai, doosra minimum.

Numerical Example: Leaf values 3,5 => MAX chooses 5 ; 2,9 => MAX chooses 9 ; MIN chooses $\min(5,9)=5$.

8. Alpha-Beta Pruning

Theory (English): Alpha-beta pruning removes branches that cannot affect final decision in MiniMax.

Hindi: Jo branch answer ko affect nahi karegi use cut kar dete hain.

Numerical Example: If $\alpha=5$ and next branch $\beta=4$, then $\beta \leq \alpha$ so prune remaining branch.

9. Alpha-Beta Cut-off Advantages

Theory (English): It reduces search time, evaluates fewer nodes and improves efficiency.

Hindi: Kam nodes check karne padte hain isliye fast hota hai.

Example: Instead of checking all 8 leaves, only 5 leaves may be checked.